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Roll No

CS-7102 (GS)

B.E. VII Semester Examination, June 2020

Grading System (GS)

Simulation and Modelling

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Explain the different step involved in a simulation study.
 b) Discuss the advantage, disadvantage and pitfalls of simulation.

2. Explain the following component of simulation system with an example of bank system.
 i) System ii) Entity
 iii) Attribute iv) Activity
 v) Event

3. a) What do you mean by System Modeling? What are the steps used in modeling? Explain with an example.
 b) Write short note on dynamic physical models.

4. Differentiate between random numbers and pseudo random number now could random numbers that are uniform on the interval (0,1) be transformed into random number that are uniform on the interval (–11,17)?

5. a) Explain the Rejection method of generating non-uniform continuous distributed random numbers.
 b) What is queuing system? What are the important characteristics of a queuing system?

6. Draw Cobweb model of the following market models and determine whether market is stable or unstable.
 $Q = 12.4 – 1.2\ P$
 $S = 8.0 – 0.6\ P–1$
 $P_0 = 1.0$
 $Q = 5$

7. What is queueing model? How it is useful for simulation? Explain are different kind of queueing model in detail.

8. Write short notes on:(Any two)
 a) Cobweb model
 b) SIM SCRIPT
 c) Time Series Analysis

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